

数学物理方程(理科版)

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第三章 热传导方程

§1. 初值问题

§1.1 Fourier变换

三、Fourier变换性质

(3) 乘多项式性质: 若 $f(x), xf(x) \in L(-\infty, \infty)$, 则有:

$$\left(xf(x)\right)^{\wedge}(\lambda) = i \frac{d}{d\lambda} \hat{f}(\lambda).$$

证明: 由于 $f(x), xf(x) \in L(-\infty, \infty)$,

$$\begin{aligned} \frac{d}{d\lambda} \hat{f}(\lambda) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \frac{d}{d\lambda} \left(f(x) e^{-i\lambda x} \right) dx \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} -ixf(x) e^{-i\lambda x} dx = -i \left(xf(x)\right)^{\wedge}(\lambda), \end{aligned}$$

从而

$$\left(xf(x)\right)^{\wedge}(\lambda) = i \frac{d}{d\lambda} \hat{f}(\lambda).$$

一般当 $f(x), xf(x), \dots, x^m f(x) \in L(-\infty, \infty) (m \geq 1)$ 时有:

$$\left(x^m f(x)\right)^{\wedge}(\lambda) = i^m \frac{d^m}{d\lambda^m} \hat{f}(\lambda).$$

(4) 平移性质: 若 $f(x) \in L(-\infty, \infty), a \in \mathbb{R}$, 则有:

$$\left(f(x-a)\right)^{\wedge}(\lambda) = e^{-ia\lambda} \hat{f}(\lambda),$$

$$\left(e^{iax} f(x)\right)^{\wedge}(\lambda) = \hat{f}(\lambda - a).$$

(5) 伸缩性质: 若 $f(x) \in L(-\infty, \infty)$, $k \in \mathbb{R}, k \neq 0$, 则有:

$$(f(kx))^{\wedge}(\lambda) = \frac{1}{|k|} \hat{f}\left(\frac{\lambda}{k}\right).$$

证明: 不妨设 $k < 0$, 由定义可得:

$$\begin{aligned} (f(kx))^{\wedge}(\lambda) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(kx) e^{-i\lambda x} dx \\ &= \frac{1}{\sqrt{2\pi}} \int_{\infty}^{-\infty} f(\xi) e^{-i\lambda \frac{\xi}{k}} d\frac{\xi}{k} \\ &= -\frac{1}{\sqrt{2\pi}} \frac{1}{k} \int_{-\infty}^{\infty} f(\xi) e^{-i\xi \frac{\lambda}{k}} d\xi = \frac{1}{|k|} \hat{f}\left(\frac{\lambda}{k}\right). \end{aligned} \quad (1)$$

证毕.

(6) 对称性质 若 $f(x) \in L(-\infty, \infty)$, 则有:

$$(f(x))^{\vee}(\lambda) = (f(x))^{\wedge}(-\lambda) = (f(-x))^{\wedge}(-\lambda).$$

$$(f(x))^{\wedge}(\lambda) = (f(x))^{\vee}(-\lambda) = (f(-x))^{\vee}(-\lambda).$$

(7) 卷积 1) 定义: 若 $f(x), g(x) \in L(-\infty, \infty)$, 则定义 $f(x), g(x)$ 的卷积为

$$f * g(x) = \int_{-\infty}^{\infty} f(x-t)g(t)dt.$$

2) 性质: 若 $f(x), g(x) \in L(-\infty, \infty)$, 则 $f * g(x) \in L(-\infty, \infty)$.

证明：由Fubini定理直接计算得：

$$\begin{aligned}\int_{-\infty}^{\infty} |f * g(x)| dx &= \int_{-\infty}^{\infty} \left| \int_{-\infty}^{\infty} f(x-t)g(t)dt \right| dx \\ &\leq \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} |f(x-t)g(t)| dt dx \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} |f(x-t)g(t)| dx dt \quad (\text{Fubini THM}) \\ &= \int_{-\infty}^{\infty} |g(t)| dt \int_{-\infty}^{\infty} |f(x-t)| dx \\ &= \int_{-\infty}^{\infty} |g(t)| dt \int_{-\infty}^{\infty} |f(\xi)| d\xi \quad (x-t=\xi \text{ transform}) \\ &= \left(\int_{-\infty}^{\infty} |f(\xi)| d\xi \right) \left(\int_{-\infty}^{\infty} |g(t)| dt \right) < \infty.\end{aligned}$$

(2)

3) Fourier变换的卷积性质: 若 $f(x), g(x) \in L(-\infty, \infty)$, 则

$$(f * g(x))^{\wedge}(\lambda) = \sqrt{2\pi} \hat{f}(\lambda) \hat{g}(\lambda).$$

证明: 利用Fubini定理直接计算得:

$$\begin{aligned} (f * g(x))^{\wedge}(\lambda) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f * g(x) e^{-i\lambda x} dx \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \left(\int_{-\infty}^{\infty} f(x-t) g(t) dt \right) e^{-i\lambda x} dx \end{aligned}$$

$$\begin{aligned}
&= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \left(\int_{-\infty}^{\infty} f(x-t)e^{-i\lambda x} g(t) dx \right) dt \quad (\text{Fubini THM}) \\
&= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \left(\int_{-\infty}^{\infty} f(x-t)e^{-i\lambda(x-t)} dx \right) g(t)e^{-i\lambda t} dt \\
&= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \left(\int_{-\infty}^{\infty} f(\xi)e^{-i\lambda\xi} d\xi \right) g(t)e^{-i\lambda t} dt \quad x-t=\xi \\
&= \frac{1}{\sqrt{2\pi}} \left(\int_{-\infty}^{\infty} f(\xi)e^{-i\lambda\xi} d\xi \right) \left(\int_{-\infty}^{\infty} g(t)e^{-i\lambda t} dt \right) \\
&= \sqrt{2\pi} \hat{f}(\lambda) \hat{g}(\lambda).
\end{aligned}$$

证毕.

例题：

例1： 设

$$f_1(x) = \begin{cases} 1, & |x| \leq A, \\ 0, & |x| > A, \end{cases}$$

求 $\hat{f}_1(\lambda)$.

解：由定义

$$\hat{f}_1(\lambda) = \frac{1}{\sqrt{2\pi}} \int_{-A}^A e^{-i\lambda x} dx = \sqrt{\frac{2}{\pi}} \frac{\sin \lambda A}{\lambda}.$$

例2: 设

$$f_2(x) = \begin{cases} e^{-x}, & x \geq 0, \\ 0, & x < 0, \end{cases}$$

求 $\hat{f}_2(\lambda)$.

解: 由定义

$$\hat{f}_2(\lambda) = \frac{1}{\sqrt{2\pi}} \int_0^{\infty} e^{-x} e^{-i\lambda x} dx = \frac{1}{\sqrt{2\pi}} \frac{1}{1+i\lambda}.$$

例3: 设 $f_3(x) = e^{-|x|}$, 求 $\hat{f}_3(\lambda)$.

解: 由于 $f_3(x) = f_2(x) + f_2(-x)$, 由线性性质和伸缩性质, 得:

$$\begin{aligned}\hat{f}_3(\lambda) &= \hat{f}_2(\lambda) + [f_2(-x)]^\wedge \\ &= \hat{f}_2(\lambda) + \hat{f}_2(-\lambda) \\ &= \frac{1}{\sqrt{2\pi}} \frac{1}{1+i\lambda} + \frac{1}{\sqrt{2\pi}} \frac{1}{1-i\lambda} \\ &= \frac{1}{\sqrt{2\pi}} \frac{2}{1+\lambda^2}.\end{aligned}$$

例4: 设 $f_4(x) = e^{-x^2}$, 求 $\hat{f}_4(\lambda)$.

解: 由定义及分部积分公式

$$\begin{aligned}\hat{f}_4(\lambda) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-x^2} e^{-i\lambda x} dx \\&= \frac{1}{\sqrt{2\pi}} \frac{i}{\lambda} \left[e^{-x^2} e^{-i\lambda x} \Big|_{-\infty}^{\infty} + 2 \int_{-\infty}^{\infty} x e^{-x^2} e^{-i\lambda x} dx \right] \\&= \frac{2i}{\lambda} \left[x e^{-x^2} \right]^{\wedge} = \frac{2i}{\lambda} \left[i \frac{d}{d\lambda} \hat{f}_4(\lambda) \right] = -\frac{2}{\lambda} \frac{d\hat{f}_4(\lambda)}{d\lambda}. \\ \hat{f}_4(0) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-x^2} dx = \frac{1}{\sqrt{2}}.\end{aligned}$$

所以

$$\begin{cases} \frac{d\hat{f}_4(\lambda)}{d\lambda} = -\frac{\lambda}{2}\hat{f}_4(\lambda), \\ \hat{f}_4(0) = \frac{1}{\sqrt{2}}. \end{cases}$$

解之得：

$$\hat{f}_4(\lambda) = \frac{1}{\sqrt{2}}e^{-\frac{\lambda^2}{4}}.$$

例5: 设 $f_5(x) = e^{-Ax^2}$ ($A > 0$), 求 $\hat{f}_5(\lambda)$.

解: 由伸缩性质

$$\begin{aligned}\hat{f}_5(\lambda) &= \left[f_4(\sqrt{A}x) \right]^\wedge(\lambda) \\ &= \frac{1}{\sqrt{A}} \hat{f}_4\left(\frac{\lambda}{\sqrt{A}}\right) \\ &= \frac{1}{\sqrt{2A}} e^{-\frac{\lambda^2}{4A}}.\end{aligned}\tag{3}$$

同学们，再见！